Unit 9: Worked Examples

1. Find the solution(s) to $x^2 - 41 = -5$ by using square roots.

$$x^2 - \cancel{41} = -5$$
$$\cancel{+41} \quad +41$$
$$\sqrt{x^2} = \sqrt{36}$$
$$x = \pm 6$$

2. Solve $x^2 + 8x + 3 = -4$ by completing the square.

$$x^2 + 8x + 3 = -4$$
$$+4 \quad +4$$
$$x^2 + 8x + 7 = 0$$
$$4$$
$$x^2 + 8x + 7 - 16$$
$$x^2 + 8x - 9$$

$$x^2 + 8x - \cancel{9} = 0$$
$$+9 \quad +9$$
$$x^2 + 8x = 9$$
$$\sqrt{(x+4)^2} = \sqrt{9}$$
$$x + 4 = \pm\sqrt{9}$$
$$x = -1 \text{ or } x = -7$$

3. Solve $12x^2 + 8x - 15 = 0$ by using the quadratic formula.

$A = 12$

$B = 8$

$C = -15$

$$\frac{-8 \pm \sqrt{8^2 - 4(12)(-15)}}{2(12)}$$

$$\frac{-8 \pm \sqrt{64 + 720}}{24}$$

$$\frac{-8 \pm \sqrt{784}}{24}$$

$$\frac{-8 + \sqrt{784}}{24} \quad \frac{-8 - \sqrt{784}}{24}$$

$$x = \frac{5}{6} \quad x = -\frac{3}{2}$$